

Management Information System

Unit - 4

Development of MIS Plan and Contents of MIS plan

Introduction:

- Like any other business activity, the long range plan for development as well as implementation is a basic necessity of MIS.
- In MIS, information is recognized as a major resource like capital, time and capacity.
- If the resource is to be managed well, it calls upon the management to plan for it and control it for the appropriate use in the organization.
- Computer technology has now become user-friendly in the manipulation of Information function of Storage, Communications, Intelligence and Language.
- Computers can now be used as a tool for information processing and communication. It can now be used for storing large database or knowledge base. It can be used for knowing the current status of any aspect of the business due to its on-line real time processing capability.
- Such a broad-base activity can be executed only when it is conceived as a system.
 - ✓ This system should deal with management information and not with data processing alone.
 - ✓ It should provide support for management planning, decision making and action.
 - ✓ It should support the needs of the lower management as well as that of the top management.
 - ✓ It should provide support to the changing needs of business management.
- The plan of the MIS is concurrent to the Business Plan of the organization.
- To ensure that the information needs is concurrent with the implementation of the business plan, it is necessary that the business plan states the information needs.
- The information needs are then traced to the source data and the systems in organization which generate such a data.
- The details of information are provided to the operations management to assess the status of an activity and to find ways to make up, if necessary.
- Once the management needs are translated into information needs, it is left for the designer to evolve a plan of development and implementation.

Contents of MIS plan

The MIS plan would have the following contents which would be dealt by the designer under a support from the top management.

1. MIS goals and objective
2. Strategy for plan achievement
3. The architecture of MIS
4. The system development schedule
5. Hardware and software plan

1. **MIS goals and objective**

- It is necessary to develop the goals and objectives for the MIS which will support the business goals. The MIS goals and objectives will consider management philosophy, policy constraints, business risks, internal and external environment of the organization.
- The typical statements of the goals are :
 - ✓ Provide online information on the stocks, markets and the accounts balances.
 - ✓ The query processing should not exceed more than three seconds.
 - ✓ The focus of the system will be on the end user computing and access facilities.
- Such statements of the goals and objectives enable the designer to set the direction and design implementation strategies for the MIS plan.

2. **Strategy for the plan achievement**

The designer has to take a number of strategic decisions for the achievement of the MIS goals and objectives.

- a) Development Strategy: An online, a batch, a real time
- b) System Development Strategy: An approach to the system development – Operational versus Functional; database versus Conventional approach; Distributed versus Decentralized processing
- c) Resource for system development: In-house versus external, customized development versus the use of packages.
- d) Manpower Composition: Analyst, programmer skills.

3. **The architecture of the MIS**

- The architecture of the MIS plan provides a system and subsystem structure and their input, output and linkages.
- It also provides a way to handle the systems or subsystems by way of simplification, coupling and decoupling of subsystems.

- It indicates the details of the subsystems from the data entry to processing, analysis to modeling, and storage to printing.

4. **System Development Schedule**

- A schedule is made for the development of the system. While preparing the schedule due consideration is given to the importance of the system in overall information requirement.
- The development schedule is to be weighed against the time scale for achieving a certain information requirement linked to a business plan. If these are not fully met, it is necessary to revise the time schedule and also the development schedule, whenever necessary.

5. **Hardware and software plan**

Giving importance to the technical and operational feasibility, the economics of investment is

- Then the plan of procurement is made after selecting the hardware and software.
- One can take the phased approach of investment starting from the lower configuration of hardware going over to higher as development takes place.
- The process is to match the technical decisions with the financial decisions.
- The system development schedule is linked with the information requirements which in turn, are linked with the goals and objectives of the business.

The organizations which do not care to take proper decisions in these areas suffer from over- investment, under-utilization and are not able to meet the critical information requirements.

It is important to note the following points:

1. The organization's strategic plan should be the basis for the MIS strategic plan.
2. The information system development schedule should match with the implementation schedule of the business plan.
3. The choice of information technology is a strategic business decision and not a financial decision.

MIS Plan is Linked to the Business Plan

MIS goals and objectives: it is necessary to develop the goals and objectives for the MIS which will support the business goals. The MIS goals and objectives will consider management philosophy, policy constraints, business risks, internal and external environment of the organization and the business.

The goals and the objectives of the MIS would be so stated that they can be measured.

The typical statements of the goals are as under:

- Provide on-line information on the stocks, markets and the accounts balances.
- The query processing should not exceed more than three seconds.
- The focus of the system will be on the end user computing and access facilities.
- Information support will be the first in the strategic areas of management such as marketing or service or technology.

Table: Business Plan versus MIS Plan

Business	MIS Plan
Business goals and objectives.	Management information system, objectives, consistent to the business goals and objectives.
Business plan and strategy	Information strategy for the business plan implementation playing a supportive role.
Strategy planning and decisions.	Architecture of the Management Information system to support decisions.
Management Plan for execution and control.	System development schedule, matching the plan execution.
Operation plan for the execution.	Hardware and software plan for the procurement and the implementation.

Such statements of the goals and objectives enable the designer to set the direction and design implementation strategies for the MIS.

Strategy for the plan achievement: The designer has to take a number of strategic decisions for the achievement of the MIS goals and objectives. They are:

- a) Development strategy: An online, a batch, a real time.
- b) System Development Strategy: an approach to the system development – Operational versus Functional; Accounting versus Analysis; Database versus Conventional Approach; Distributed versus Decentralized processing ; one Database versus Multiple database SSAD vs. OOT.
- c) Resource for System Development: In – house versus external, customized development versus the use of packages.

Classification of information – organizational, functional, knowledge, decision support and operational

Ascertaining the class of information

- Ascertaining the information needs of the management of the business execution is a complex task. The complexity can be handled if the information is classified on the basis of its application and the user, which becomes the basis for the ascertainment.

The various classes of Information are as follows:

- It is the information required by a number of personnel, departments and divisions or the functions in the organization. Such information can be determined by constructing a matrix of information versus user

Matrix of Information versus User for a Personnel Function

Information entry	Manager (Personal)	Manager (Production)	Manager (Administration)	Manager (Accounts)
Employees attendance	X	X	X	X
Salary wages and overtime	X	X	X	X
Human resources information	X	X	X	X

- It can be observed from the table that the information entity is one, but its usages are different.
- For example, the employee attendance information would be used by the personnel department for legal compliance of maintaining the muster recommended by the Factory Inspector.
- The production manager would use it for scheduling, rescheduling and loading of the jobs on the shop floor depending upon the persons present.
- The corporate planning and administration will use it for manpower assessment and control and manpower forecasting.
- It is necessary to map the information in terms of the data source, generation and usage, so that the designer can provide a path from acquisition to the generation and storage.

- Since the usage of the organizational information is at different levels for different purposes, it is advisable to store the data in the form of the database which will be used by the users for generating their respective information needs.

Function Information

- The functional information is defined as a set of information required by the functional head in conducting the administration and management of the function.
- This information is purely local to that function and by definition, does not have a use elsewhere. This information is used by a manager to plan and control his function.
- Function information is largely factual, statistical and detailed in multi-dimensions of the function. For example, if you take the sales information, it can be processed in several ways, viz. the product, the product groups, the market segment, the geographic zones, the locations, the customer.
- The functional information is normally generated at equal time intervals, say monthly quarterly etc. for understanding the trend and making comparisons against the time scale. Such information is used for planning, budgeting and controlling the operations of the function.
- The functional information can be accessed on the following three parameters – the work design, the responsibility and the functional objectives.

➤ Work Design

For example, for the customer order scrutiny the available stock, the price, the terms of payment and the probable delivery is an information set evolved out of the work design of customer order processing. The procedure of the order processing requires this information.

➤ Responsibility

- The managers in the functional areas of management are responsible for achieving the targets and accomplishing the goals and objectives.
- It is therefore necessary to inform and update the information on targets at regular intervals to enable him to make or change decisions in his domain of operations.
- For example, the marketing manager has monthly target of Rs. 1 million order booking. Since he is responsible for achieving the targets, it would be necessary to inform him on these aspects at regular intervals.

➤ **Functional Objectives**

- Each function has its own objective which is derived out of the corporate goals.
- The functional goals and objectives are necessary to achieve overall corporate achievements.
- In summary, the functional information would emanate from the work design and procedures, the managerial responsibility accounting, and with reference to the functional goals and objectives.
- The information that measures the business activity and evaluates the performance on the key target areas, is the functional information.

➤ **Knowledge information**

- The knowledge information creates an awareness of those aspects of business where the manager is forced to think, decide and act. Such an information shows the trend of the activity, or a result against the timescale.
- For example, whether the sales are declining and the trend is likely to continue in the next quarter. The product is failing continuously on one aspect and the reason of failure is the process of manufacturing.
- Such information pin-points the area or entity and forces the manager to act. It highlights the deviations from the norm or standard and also any abnormal developments which are not in congruence with the forecasts or expectations.
- Such information gives rise to business decisions, which will affect the process of business
- Knowledge information is tracked continuously reported in a fixed format, for consistency and at fixed intervals for updating the knowledge base.

➤ **Decision Support Information**

- Most of the information required by the middle and the top management is for decision making. The information does not act as a direct input to the decision making procedure or formula but supports the manager in the effort of decision making.
- The support may act two ways, one for justifying the need of a decision, and the other as an aid to decision making.
- The source of this information could be internal or external to the organization. It can be determined by identifying tools, techniques, models and procedures, used by the managers in the decision making.

➤ **Operational Information**

- This information is required by the operational and the lower levels of the management.
- The main purpose of this information is fact finding and taking such actions or decisions which will affect the operations at a micro-level.
- The decisions may be to stay on overtime, draw additional material, change the job from one machine to the other, send a reminder to the supplier for the supply of material.
- These decisions are such that they make the routine administration of the business smooth and efficient.
- The sources of information are largely internal through transaction processing.

Management of Quality in the MIS

Management of Quality in MIS

Management of Quality in Management Information Systems (MIS) means **maintaining high accuracy, reliability, and usefulness of information in the system**. Good quality information helps managers make **correct decisions and improve organizational performance**.

Simply, **quality management in MIS ensures that the information produced by the system is correct, complete, timely, and useful**.

Definition:

Management of Quality in MIS is the process of ensuring that information generated by the system is accurate, complete, timely, consistent, and relevant so that managers can make effective decisions.

Main Elements/ Aspects/ Components/ Features of Quality in MIS

1. Accuracy

Meaning: The information should be **correct and error-free**.

Example:

If a sales system shows **₹10,000 sales instead of ₹1,000**, it creates wrong reports. MIS must ensure **accurate data entry and processing**.

2. Completeness

Meaning: The information should include **all necessary details**.

Example:

A customer database should contain **name, phone number, address, and purchase history**. Missing details make information less useful.

3. Timeliness

Meaning: Information should be **available at the right time**.

Example:

A manager needs **daily sales reports** to decide how much stock to order. If the report comes after a week, it is not useful.

4. Consistency

Meaning: The same data should be **uniform across different systems or reports**.

Example:

If the inventory system shows **50 units in stock** but the sales system shows **40 units**, it creates confusion. MIS must keep data consistent.

5. Relevance

Meaning: Information should be **useful for the decision being made**.

Example:

A marketing manager planning a campaign needs **customer buying patterns**, not employee attendance records.

Example of Quality Management in MIS

In a **retail store MIS**:

- Barcode scanners ensure **accurate sales data**.
- The system updates stock **immediately (timely)**.
- Customer details are stored **completely**.
- All departments see the **same inventory data (consistent)**.

This helps managers **track sales, manage inventory, and make better business decisions**.

- Information quality is required to be very high, since it is a corporate resource.
- The quality of the information is the result of the quality of the input data, processing design, system design, system and procedures which generate such a data, and the management of the data processing function.
- It is possible to measure the quality of information on certain parameters.

Quality Parameters

- ✓ Complete data of all the transactions
- ✓ Valid transaction and input data
- ✓ Accuracy and precision
- ✓ Relevance to the user
- ✓ Timely information
- ✓ Meaningful and complete information

- The quality of these important parameters is ensured by conducting a proper systems analysis, designing suitable computer system and ensuring its maintenance from time to time, and also subjecting it to audit checks to ensure the system integrity.

The quality of the parameters is assured if the following steps are taken.

- All the input is processed and controlled.
- All updating and corrections are completed before the data processing begins.
- Inputs (transactions, documents, fields and records) are subjected to validity checks.
- The access to the data files is protected and secured through an authorization scheme.
- Intermediate processing checks are introduced to ensure that the complete data is processed right through i.e., run to run controls.
- Due attention is given to the proper file selection terms of data, periods, and so on.
- Back up of the data and files are taken to safeguard corruption or loss of data.
- The system audit is conducted from time to time to ensure that the computer system specifications are not violated.
- The system modifications are approved by following a set procedure which begin with authorization of a change to its implementation followed by an audit.
- Systems are developed with a standard specification of design and development.
- Computer system processing is controlled through program control, process control and access control.

Information is a corporate resource as important as the capital labor, know-how, etc. and is being used for decision making. Its' quality, therefore is required to be very high. Low quality information would adversely affect the organizational performance as it affects decision making. The quality of information is the result of quality of input data, processing design, system design, system procedure which generate such a data, and the management of the data processing function. Quality, unlike any other product is not an absolute concept. Its level is determined with reference to the context and its use, and the user. Perfect quality just as perfect information is non- achievable and has cost benefit implications.

However, it is possible to measure the quality of information on certain parameters. All these parameters need not have a very high value in terms of the unit of measure. Some parameters may have lesser importance in the total value on account of their relevance in the information and its use.